# 30-METER HEAVY-DUTY POLYGONAL TUBULAR STEEL MONOPOLE

## DETAILED DESIGN DRAWING (DDD) ENGINEERING & STRUCTURAL CALCULATION MANUAL

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## 1. Executive Summary & Design Basis

This Engineering Specification & Structural Calculation Manual serves as the technical companion to the 30-Sheet Detailed Design Drawing (DDD) Dossier (30M_315KPH_150KPA_G1_TelTech-FWA DDD.pdf). It sets forth the definitive civil, structural, mechanical, electrical, and geotechnical criteria governing the nationwide procurement, fabrication, foundation construction, and site erection of standard 30-meter heavy-duty polygonal tubular steel monopoles for TelTech's Nationwide 5G Standalone (SA) Fixed Wireless Access (FWA) Expansion Program.

The engineering architecture addresses the most extreme typhoon exposure corridors in the Philippine archipelago (NSCP 2015 7th Edition Zone 1 basic wind speed of **315 km/h** [87.5 m/s], 3-second gust), coupled with a standard allowable soil bearing capacity floor of **qa >= 150 kPa** (with pre-engineered soft-soil remediation protocols down to 100 kPa). The facility is fully optimized for co-location of high-capacity 5G Band n50 macro active antenna systems, integrated 8T8R remote radio units, Starlink satellite backhaul terminals, and future microwave transmissions.

## 2. Monopole Structural Geometry & Material Specifications

The 30-meter monopole is engineered as a 12-sided polygonal tapered tubular steel column fabricated in five (5) slip-joint telescoping sections with a constant diametrical taper ratio of 15 mm per meter of height. All sections are joined via precision male/female tapered friction slip joints having a minimum overlap length of 1.5 times the internal sleeve diameter (1,500 mm).

| SECTION | ELEVATION RANGE | SECTION LENGTH | BASE O.D. | TOP O.D. | WALL THK | STEEL GRADE | WEIGHT (kg) |
|---|---|---|---|---|---|---|---|
| **Section S1 (Base)** | 0.00m - 6.00m | 6,000 mm | 1,050 mm | 960 mm | 14 mm | ASTM A572 Gr 50 | 2,240 kg |
| **Section S2** | 4.50m - 10.50m | 6,000 mm | 982 mm | 892 mm | 12 mm | ASTM A572 Gr 50 | 1,780 kg |
| **Section S3** | 9.00m - 15.00m | 6,000 mm | 914 mm | 824 mm | 12 mm | ASTM A572 Gr 50 | 1,650 kg |
| **Section S4** | 13.50m - 19.50m | 6,000 mm | 846 mm | 756 mm | 8 mm | ASTM A572 Gr 50 | 1,020 kg |
| **Section S5 (Top)** | 18.00m - 30.00m | 12,000 mm | 778 mm | 450 mm | 6 mm | ASTM A572 Gr 50 | 1,180 kg |
| **TOTAL SHAFT** | **0.00m - 30.00m** | **30.0 Meters** | **1,050 mm** | **450 mm** | **6-14 mm** | **Fy = 345 MPa** | **7,870 kg** |

## 3. Wind Loading & Structural Reaction Formulations

Wind pressure calculations conform strictly to NSCP 2015 Section 207 and ANSI/TIA-222-H. The velocity pressure *qz* is evaluated using the governing expression:

**qz = 0.613 * Kz * Kzt * Kd * Ke * V^2 * Iw (N/m2)**

where *V = 87.5 m/s* (315 km/h for Zone 1), *Exposure Category C* (open rolling terrain), *Kz = 1.18* at 30m, *Kzt = 1.00*, *Kd = 0.95* (polygonal tubular shaft), *Ke = 1.00*, and *Iw = 1.15* (essential telecom facility).

| LOAD COMBINATION / LIMIT STATE | OVERTURNING MOMENT (Mu) | BASE SHEAR (Vu) | AXIAL GRAVITY (Pu) | TIP DEFLECTION (Delta) | ANTENNA TILT (Theta) |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| **ULS: 1.2D + 1.0W (315 km/h Extreme)** | 1,850.0 kN·m | 145.0 kN | 120.0 kN | 380 mm (Buckling Check) | 1.85 deg (Plastic Yield Check) |
| **ULS: 0.9D + 1.0W (Max Net Uplift)** | 1,825.0 kN·m | 145.0 kN | 85.0 kN | 375 mm | 1.82 deg |
| **SLS: 1.0D + 1.0W (180 km/h Operating)** | 600.0 kN·m | 48.0 kN | 100.0 kN | **185 mm (<= 300mm)** | **0.62 deg (<= 1.00 deg)** |
| **Seismic ULS: 1.2D + 1.0E (Zone 4)** | 420.0 kN·m | 38.0 kN | 125.0 kN | 75 mm | 0.24 deg |

## 4. Foundation Mat & Pedestal Geotechnical Stability

The standard foundation comprises an isolated reinforced concrete pad-and-chimney mat footing. Sizing parameters:
• **Mat Footing:** 5.00m x 5.00m x 0.80m thickness (20.0 m3 concrete volume)
• **Pedestal Chimney:** 1.80m x 1.80m x 1.20m height (+0.30m above NGL, -0.90m below NGL)
• **Total Founding Depth:** -1.70m below finished grade level on competent undisturbed soil.
• **Blinding Bed:** 100mm lean concrete (f'c >= 15 MPa) cast immediately after excavation.
• **Structural Concrete:** Class 'A' concrete with minimum f'c >= 28 MPa (4,000 psi) at 28 days.
• **Reinforcement:** Mat bottom grid: 20mm dia. @ 150mm c/c E.W.; Mat top grid: 16mm dia. @ 150mm c/c E.W.; Pedestal verticals: 20-25mm dia. with 90° hooked anchor legs into the bottom mat; Pedestal ties: 12mm dia. closed stirrups @ 100mm c/c upper zone, 150mm c/c remainder.

| STABILITY CRITERIA | DESIGN FORMULATION | ALLOWABLE LIMIT | CALCULATED VALUE | FACTOR OF SAFETY | STATUS |
|---|---|---|---|---|---|
| **Max Soil Pressure (qmax)** | q = (W/A) * [1 + 6e/B] | qa <= 150.0 kPa | 118.4 kPa | **SF = 2.45 (>= 2.0)** | **PASS** |
| **Min Soil Pressure (qmin)** | q = (W/A) * [1 - 6e/B] | qmin >= 0 (No Uplift) | 12.2 kPa (Compression) | 100% Bearing Area | **PASS** |
| **Overturning Stability** | SF_ot = M_resist / M_ot | SF >= 1.50 | M_res = 3,975 kN·m | **SF = 2.15 (>= 1.50)** | **PASS** |
| **Sliding Resistance** | SF_sl = (mu*W + Pp) / V | SF >= 1.50 (mu = 0.35) | R_slide = 268 kN | **SF = 1.85 (>= 1.50)** | **PASS** |

## 5. Baseplate & Hold-Down Anchor Bolt Assembly

• **Baseplate:** 1,400 mm outside diameter x 60 mm thick solid steel plate (ASTM A572 Grade 50, Fy = 345 MPa). Incorporates a central 400 mm diameter cable feed port and twelve (12) radial gusset stiffeners (25 mm thick x 250 mm height, Grade 50) welded with complete joint penetration (CJP) bevel welds.
• **Anchor Bolts:** Twelve (12) M36 Grade 105 (ASTM F1554) high-strength all-thread anchor rods arranged on a 1,200 mm pitch circle diameter (PCD). Total bolt length is 1,800 mm with an effective embedment depth of 1,450 mm in the pedestal/mat concrete. Each rod is fitted with an ASTM A36 bottom anchor plate (150mm x 150mm x 20mm thk) locked with tack-welded heavy hex nuts.
• **Grout Bed:** 50 mm thick non-shrink high-strength cementitious grout bed (f'g >= 50 MPa) placed between the pedestal concrete and the underside of the baseplate after optical leveling.

## 6. Electrical, Hybrid Power & Grounding Grid Specifications

• **Utility Service Drop:** 230V 1-Phase 60Hz drop from local distribution co-op, terminated at a +3.5m SHS metering pedestal with NEMA 3R revenue kWh meter socket and 63A 2-pole main circuit breaker.
• **Manual Transfer Switch (MTS):** 63A 2-pole break-before-make switch with external 63A emergency generator receptacle.
• **AC Power Distribution Board (ACPDB):** NEMA 3R enclosure with Class I+II surge protective device (SPD, Imax = 40kA, Up <= 1.5kV) and branch breakers.
• **DC Modular Power Plant:** 12kW high-efficiency (-48V DC floating output) telecom rectifier system with 3x 4kW hot-swappable modules (N+1 redundancy).
• **Energy Storage:** Two (2) 48V 100Ah Lithium Iron Phosphate (LiFePO4) battery strings (9.6 kWh total capacity) delivering 10-14 hours of autonomous site backup.
• **Site Earthing Grid:** Buried closed loop of 50 mm2 bare stranded soft-drawn copper conductor buried at -800 mm depth around the 8m x 8m compound perimeter, bonded to four (4) 19mm dia. x 3.0m copper-clad steel ground rods housed in precast concrete inspection chambers with removable disconnect test links. Tower base connected via four (4) symmetrical Cadweld molecular exothermic welds. Combined earth resistance shall not exceed **5.0 Ohms**.

## 7. Master Drawing Register Cross-Reference Index

The table below itemizes the complete 30-sheet drawing dossier contained in 30M_315KPH_150KPA_G1_TelTech-FWA DDD.pdf:

| SHEET NO. | DRAWING NO. | SHEET TITLE | SCALE | DISCIPLINE |
|---|---|---|---|---|
| **Sheet 01 of 30** | **TT_AR_000** | COVER PAGE & MASTER DRAWING INDEX | 1:1 | Architectural (General) |
| **Sheet 02 of 30** | **TT_AR_001** | DESIGN PARAMETERS, MATERIAL SPECS & 3D PERSPECTIVE VIEW | AS SHOWN | Architectural (General) |
| **Sheet 03 of 30** | **TT_AR_002** | PHILIPPINE BASIC WIND LOAD REGIMES & STRUCTURAL LIMIT STATES | NTS | Architectural (General) |
| **Sheet 04 of 30** | **TT_AR_003** | SITE LOCATION MAP, SEARCH RING & SITE DEVELOPMENT PLAN | 1:500 | Architectural (General) |
| **Sheet 05 of 30** | **TT_AR_004** | STANDARD 8.0M x 8.0M COMPOUND PHYSICAL LAYOUT PLAN | 1:100 | Architectural (General) |
| **Sheet 06 of 30** | **TT_AR_005** | 30-METER MONOPOLE TOWER & COMPOUND ELEVATION VIEW | 1:100 | Architectural (General) |
| **Sheet 07 of 30** | **TT_ST_001** | MONOPOLE KEY PLAN, SECTIONAL GEOMETRY & STRUCTURAL REACTIONS | 1:50 | Structural / Civil |
| **Sheet 08 of 30** | **TT_ST_002** | PAD AND CHIMNEY MAT FOUNDATION STRUCTURAL DETAILS | 1:50 | Structural / Civil |
| **Sheet 09 of 30** | **TT_ST_003** | FOUNDATION REINFORCEMENT & REBAR SECTIONAL ARRANGEMENT | 1:25 | Structural / Civil |
| **Sheet 10 of 30** | **TT_ST_004** | FOUNDATION REBAR BENDING SCHEDULE & MATERIAL TAKE-OFF | NTS | Structural / Civil |
| **Sheet 11 of 30** | **TT_ST_005** | BASEPLATE TEMPLATE & HOLD-DOWN ANCHOR BOLT ASSEMBLY | 1:20 | Structural / Civil |
| **Sheet 12 of 30** | **TT_ST_006** | GENERAL CONCRETE SPECIFICATIONS & CIVIL CONSTRUCTION NOTES | NTS | Structural / Civil |
| **Sheet 13 of 30** | **TT_ST_007** | STRUCTURAL STEEL FABRICATION, WELDING & GALVANIZING NOTES | NTS | Structural / Civil |
| **Sheet 14 of 30** | **TT_ST_008** | OUTDOOR EQUIPMENT PAD PLINTH DETAILS (2.0M x 2.0M) | 1:25 | Structural / Civil |
| **Sheet 15 of 30** | **TT_ST_010** | ELEVATED CABLE GANTRY & CABLE LADDER STRUCTURAL DETAILS | 1:20 | Structural / Civil |
| **Sheet 16 of 30** | **TT_ST_011** | COMPOUND SECURITY FENCING DETAILS (CYCLONE MESH & RAZOR WIRE) | 1:30 | Structural / Civil |
| **Sheet 17 of 30** | **TT_ST_012** | COMPOUND ENTRANCE SWING GATE & SECURITY LOCKING ASSEMBLY | 1:25 | Structural / Civil |

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|---|---|---|---|---|
| **Sheet 18 of 30** | **TT_ST_013** | FENCE CORNER POST STIFFENERS & REINFORCED GRADE CURB DETAILS | 1:20 | Structural / Civil |
| **Sheet 19 of 30** | **TT_ST_014** | SERVICE ENTRANCE PEDESTAL & 230V UTILITY METERING POST | 1:25 | Structural / Civil |
| **Sheet 20 of 30** | **TT_ST_015** | PERIMETER SECURITY LIGHTING & FIRE EXTINGUISHER ENCLOSURE | 1:20 | Structural / Civil |
| **Sheet 21 of 30** | **TT_ST_017** | BAND n50 CROSS-POL MACRO ANTENNA MOUNTING BRACKET (30M AGL) | 1:20 | Structural / Civil |
| **Sheet 22 of 30** | **TT_ST_018** | STARLINK SATELLITE (10M AGL) & FUTURE EXPANSION MOUNTING | 1:25 | Structural / Civil |
| **Sheet 23 of 30** | **TT_EL_001** | ELECTRICAL SINGLE LINE DIAGRAM (SLD) & POWER DISTRIBUTION | NTS | Electrical / Earthing |
| **Sheet 24 of 30** | **TT_EL_002** | SITE ELECTRICAL LAYOUT, TRENCHING & CONDUIT ROUTING | 1:100 | Electrical / Earthing |
| **Sheet 25 of 30** | **TT_EL_003** | SITE EARTHING & GROUNDING GRID LAYOUT PLAN (<= 5.0 OHMS) | 1:100 | Electrical / Earthing |
| **Sheet 26 of 30** | **TT_EL_004** | MAIN GROUNDING BUSBAR (MGB) & EARTH PIT CONNECTION DETAILS | 1:10 | Electrical / Earthing |
| **Sheet 27 of 30** | **TT_EL_005** | GROUND TEST PIT & INSPECTION CHAMBER DETAILS | 1:15 | Electrical / Earthing |
| **Sheet 28 of 30** | **TT_EL_006** | TOWER LIGHTNING DOWN-CONDUCTOR & CADWELD EXOTHERMIC WELDS | 1:20 | Electrical / Earthing |
| **Sheet 29 of 30** | **TT_EL_008** | FIBER OPTIC LEAD-IN, DUCTING & UNDERGROUND PULL-BOX DETAILS | 1:20 | Electrical / Earthing |
| **Sheet 30 of 30** | **TT_ST_019** | GEOTECHNICAL SOIL INVESTIGATION & BEARING CAPACITY VERIFICATION | NTS | Structural / Civil |

## 8. Engineering Sign-Off & Regulatory Certification

| PREPARED & ENDORSED BY: | CHECKED & STRUCTURALLY CERTIFIED BY: |
|---|---|
| **OLIVER TUNGOL**<br>Lead Technical Architect & FWA Program Director<br>Telecommunications Technology Solutions Inc. (TelTech)<br>Date: 26-OCT-2026 | **JOMAR DOGILLO, PE**<br>Professional Structural Engineer (PRC Lic. No. Verified)<br>Structural Tower & Foundation Consultant<br>Date: 26-OCT-2026 |